ALESSANDRO VALENTE

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EDUCATION

2023–Present: **B.S. in Aerospace Engineering Sciences**, University of Colorado Boulder (expected May 2027)

College: College of Engineering and Applied Science
Programme: 4-year B.S., ABET-accredited
Minor: Engineering Management
Relevant Courses: ASEN 3700: Orbital Mechanics/Attitude Dynamics and Control, ASEN 3300: Aerospace Electronics and Communications, ASEN 3713: Thermodynamics and Heat Transfer, ASEN 3712: Structures, ASEN 2704: Aerospace Vehicle Design and Performance, ASEN 4013: Foundations of Propulsion (in progress)

2021–2023: **A-Levels**, St John the Baptist School, Woking, United Kingdom

Subjects: Mathematics, Physics, Computer Science

RESEARCH INTERESTS

1. Adaptive assistance tools for astronaut crews on long-duration missions without real-time ground support: automation that adjusts to each crew member's state in a way crews trust.
2. Crew state estimation from wearable ECG, EDA and EEG and from behaviour: workload, trust, situation awareness and disorientation, and when an estimate is reliable enough for an assistive tool to act on.
3. Individual differences in response to spaceflight stressors: machine learning that tailors countermeasures for spatial disorientation and communication delays to each crew member.
4. Human adaptation to spaceflight, especially modelling how vestibular function and spatial orientation perception change in altered gravity and after gravity transitions.
5. Multi-agent reinforcement learning for agents that adapt to each other, such as an assistant and the crew member it supports, with exact linear-quadratic Nash solutions as benchmarks and guides.

RESEARCH EXPERIENCE

2026–Present: **Undergraduate Researcher**, Bioastronautics, University of Colorado Boulder

Advisor: Dr. Torin K. Clark. Studies: spatial disorientation countermeasures for piloted lunar landing (NASA Human Research Program, Grant 80NSSC23K0449), Motion Sickness, PADL, and simulated post-spaceflight locomotor dysfunction.

- Run test sessions for the ongoing Motion Sickness and PADL studies, including equipment setup and data analysis; 30+ participants tested across three human-subject studies to date.
- Help conduct a galvanic vestibular stimulation study that simulates post-spaceflight locomotor dysfunction, in which participants perform functional mobility tasks and a manual control task on the Tilt Translation Sled.
- Ran testing on the Tilt Translation Sled for the spatial disorientation study (15 subjects): a real-time, model-based disorientation detector cueing a flight display eliminated boundary violations versus frequency-matched random cues, with no significant difference in secondary-task score.
- Performed a per-subject exposure–response analysis linking more triggered displays to lower piloting RMS error and fewer boundary violations, at a modest cost to secondary-task score; presented the results through CU's Summer Program for Undergraduate Research (SPUR).
- Instrument participants with wearable sensors (ECG, EDA, EEG) and data acquisition hardware, and analyse the resulting physiological datasets in MATLAB and Python.

- Designed and 3D-printed laboratory hardware, including a parametric hinged enclosure that houses a wearable physiological DAQ and clamps its electrode connector during subject runs.
- Contribute to the mechanical design of a human-rated centrifuge for altered-gravity human testing, and help integrate a Unity virtual-reality environment with the Tilt Translation Sled's real-time LabVIEW control system.

2026–Present: **Undergraduate Researcher**, Astrodynamics, University of Colorado Boulder
Advisors: Dr. Vishala Arya and Bennet Outland. Multi-agent reinforcement learning for differential games.

- Develop MADDPG.jl, a Julia package in the DifferentialGamesBase.jl ecosystem that applies multi-agent deep deterministic policy gradient (MADDPG) learning to differential games.
- Designed the package's interface, which maps a game's dynamics, costs and constraints onto a CommonRL-Interface learning environment; extending from cooperative navigation to adversarial pursuit–evasion with impassable polygon obstacles required no changes to the learning code.
- Built a cross-play evaluation showing that an apparent training plateau was a measurement artefact, and used it to show that removing redundant reward shaping in predator–prey pursuit–evasion gave stronger policies for both teams and about six times faster convergence, replicated across seeds with an effect about eight times the seed-to-seed variation.
- Implemented a stochastic linear-quadratic feedback Nash solver (coupled Riccati recursion with additive, state-multiplicative and control-multiplicative noise) and an exact equilibrium certificate (Nash gap below 10^{-11}), structured as a contribution to DifferentialGamesBase.jl.
- Validated the solver on three-satellite along-track formation assembly under Clohessy–Wiltshire–Hill dynamics: analytic expected costs matched a 2,000-run Monte Carlo simulation to within half a standard error, and the equilibrium used about 6% more propellant than centralised control.
- Verified the solver against independent computations (single-player LQR reduction, forward mean and second-moment propagation, numerical Nash gradients, certainty equivalence to machine precision), prototyping the checks in NumPy before writing the Julia.
- Build a guided policy search framework in which single-player LQR, deterministic Nash and stochastic Nash solvers are interchangeable local steps, and benchmark it against MADDPG and MAPPO on a linear-quadratic game with a known equilibrium.

MANUSCRIPTS AND ABSTRACTS

Gupta, P., Lonner, T. L., Nipunge, A. R., **Valente, A.**, & Clark, T. K. Laboratory Assessment of a Spatial Disorientation Triggering System to Aid Pilots. *Abstract submitted; full paper in preparation.*

Valente, A., Outland, B., & Arya, V. Learning Equilibria in Differential Games: Multi-Agent Reinforcement Learning with Exact Linear-Quadratic Benchmarks. *In preparation for the Journal of Open Source Software.*

ENGINEERING DESIGN EXPERIENCE

2026–Present: **Power Subsystem Lead**, ODIN Senior Design Project, University of Colorado Boulder
Advisors: Prof. Marcus Holzinger and Prof. Tomoko Matsuo. Senior Projects (ASEN 4018/4028), addressing a NASA Heliophysics priority at the intersection of orbital debris, plasma physics and space weather.

- Head the three-person power subsystem on a 23-person team designing ODIN (Orbital Debris Identification Network), a constellation of eight satellites in two dawn–dusk sun-synchronous planes at 600 km with an event-based optical payload for in-situ detection of sub-3 cm orbital debris, ending in an integrated flatsat demonstration.
- Own the EPS architecture and battery trades, power budget, power routing and PCB design, harnessing and cabling (shared with Structures), and wall-power distribution for the prototype's multi-day autonomous test.
- Designed the EPS block diagram, six control loops, and power and telemetry interfaces.

- Built an open-source MATLAB orbit and power model: two-body plus J_2 propagation (ode45 at 10^{-11} tolerances), a low-precision solar ephemeris, and conical umbra and penumbra eclipse geometry with root-solved shadow entry and exit.
- Extended the model with ground-station contacts and swept the Design Review 1 orbit across RAAN for a year, bounding the longest eclipse and the passes flown in shadow for battery and array sizing.
- Verified eclipse fractions against the analytic beta-angle expression (to 0.001) and accuracy against J_2-conserved quantities, then against MATLAB's Aerospace Toolbox: eclipse entry and exit within 1 s over 29 shadows, and yearly time in full Sun within 0.062 percentage points of full EGM2008 gravity with drag.
- Ran the battery trade across four Li-ion, Li-polymer and LFP candidates for a five-year mission, sizing each to keep at least 40% charge after the worst eclipse at end of life, and selected two COTS Li-ion packs (90 Wh) for Design Review 2.
- Simulated the five-year mission with battery ageing and the energy margin of each operating mode, and found the minimum ballistic coefficient that keeps the satellites above 600 km.

2025–Present: **Founder and Chief Engineer**, Engineering Connections Design Team, University of Colorado Boulder

- Founded and still lead a 20-member student engineering design team focused on hands-on aerospace and mechanical engineering projects.
- Direct development of a liquid-propellant sounding rocket targeting a 10 km apogee using nitrous oxide and ethanol propulsion.
- Single-handedly developed an integrated engine burn and six-degree-of-freedom (6-DOF) flight simulation in MATLAB, modelling combustion performance and full vehicle flight dynamics.
- Ported the simulation to browser-based tools so any team member can run trades: a hot-fire model (blowdown feed, combustion thermodynamics, nozzle performance) and a 6-DOF flight simulator with Monte Carlo dispersions, 1σ–3σ impact footprints, stability-margin history and sensitivity ranking.
- Rewrote the combustion solver to enforce elemental conservation, rebuilding its thermodynamic data on corrected NASA-7 coefficients with an oxygen-allocation scheme, which brought performance predictions in line with hand calculations.
- Conduct propulsion performance analysis (NASA CEA, OpenRocket), structural analysis and avionics architecture design.
- Manage project planning, design reviews and technical coordination across the propulsion, structures and recovery subsystems.

2023–Present: **Aerodynamics Team Member and Sidepod Lead**, CU Boulder Racing Team (Formula SAE)

- Design and develop aerodynamic components for the Formula SAE race car.
- Head sidepod development from concept through CAD design, analysis and manufacturing.
- Conduct CFD simulations in ANSYS to evaluate aerodynamic performance and cooling airflow.
- Perform wind tunnel testing and correlate experimental results with simulation predictions.
- Design sidepod geometry to increase downforce and improve engine cooling performance.

2025: **Project Manager**, Aerospace Vehicle Design Laboratory, University of Colorado Boulder

- Led a multidisciplinary team designing and manufacturing a 1 m wingspan boost glider.
- Managed design iterations, manufacturing planning and testing strategy.
- The final vehicle achieved one of the highest glide ranges in the cohort.

TEACHING EXPERIENCE

2026–Present: **Teaching Assistant**, EMEN 4100: Engineering Economics, University of Colorado Boulder

- Write the example answers for the course's assignments.

2025–2026: **Mathematics Instructor**, CU Upward Bound, University of Colorado Boulder

- Taught Algebra 2 to Native American high school students.
- Wrote lecture materials, homework assignments and mark schemes, and graded student work.

LEADERSHIP AND SERVICE

2025–Present: **Outreach Officer**, Society of Ethical Engineers. Organise guest lectures and discussions with government and industry professionals on engineering ethics.

2025–2026: **President**, CU Boulder Photography Club

2023–2025: **Program Coordinator**, Engineering Connections, University of Colorado Boulder. Planned and ran engineering workshops and events for first-year engineering students; developed the project-based learning activities that later evolved into the Engineering Connections Design Team.

HONOURS AND AWARDS

2026: **Colorado Engineer Magazine Scholarship**, University of Colorado Boulder

2026: **Summer Program for Undergraduate Research (SPUR)**, funded by the Engineering Excellence Fund

CERTIFICATIONS

2025: **Level 2 High-Power Rocketry Certification**, National Association of Rocketry (NAR)

2023: **Level 1 High-Power Rocketry Certification**, National Association of Rocketry (NAR)

ADDITIONAL EXPERIENCE

2022: **Formula 1 Race Simulator**, A-Level Computer Science project, St John the Baptist School

- Built a lap-by-lap race simulation in Python driven by real telemetry (FastF1), decomposing each car into high-, medium- and low-speed cornering pace and top speed, matched against each circuit's composition.
- Modelled overtaking, tyre wear by compound, pit strategy, weather and retirements, with a tkinter interface in a model–view–controller structure; moved parameter extraction offline into databases, cutting a load time of over twenty minutes.

2021: **Work Experience**, Airbus

- Gained insight into the development of the Rosalind Franklin Mars rover.
- Learned about aircraft wing design, manufacturing and analysis from Airbus engineers, and about the potential future of commercial aircraft design.

2018–2022: **Professional Motorcycle Racer**, British Superbikes and World Superbikes

- Youngest-ever competitor in the British Superbike paddock (2018, aged 12), racing in the British Junior Supersport Championship, a support class at British Superbike rounds.
- Competed in the World Superbike paddock in 2021, aged 16.
- Gained hands-on mechanical engineering experience through engine building, and designed a ram-air intake for the motorcycle.

SKILLS

Programming:	MATLAB, Python (NumPy), Julia, C++, JavaScript (React), SQL, Arduino, LaTeX
Analysis:	Orbit propagation and eclipse modelling, linear-quadratic optimal control and dynamic games, multi-agent reinforcement learning, Monte Carlo and statistical analysis, trajectory simulation, 6-DOF flight dynamics, computational fluid dynamics, finite element and structural analysis, control systems modelling, data analysis
Laboratory:	Human-subject testing, physiological instrumentation (ECG, EDA, EEG), data acquisition, 3D printing, electronics prototyping and soldering, wind tunnel testing, carbon-fibre layup and vacuum bagging, CNC machining, laser cutting, hot-wire foam cutting
Software:	SolidWorks, CATIA, Simulink, ANSYS, NASA CEA, OpenRocket, Blender, Excel
Languages:	English, Italian